

ROS2 Installation Guide

If you are using the factory image we provided, manual installation and configuration are not required.

- ROS2-Humble corresponds to Ubuntu system 22.04
- If you need to install other versions of ROS2, you only need to change the **humble** version identifier in the corresponding installation command to the desired version. This applies to all other ROS2-specific version software in the ROS2 tutorial series. Subsequent tutorials will not repeat this.

1. Set Up Locale

First, check if your locale supports UTF-8 encoding. You can run the following command to check and set UTF-8 encoding:

```
locale # Check if UTF-8 is supported

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale # Verify the settings
```

Note: The locale can be different, but it must support UTF-8 encoding.

2. Set Up Software Sources

Enable Ubuntu universe repository:

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

Add the ROS 2 apt repository to your system and authorize our GPG key with apt:

```
sudo apt update && sudo apt install curl gnupg lsb-release -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o
/usr/share/keyrings/ros-archive-keyring.gpg
```

Add the repository to the source list:

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-
keyring.gpg] http://packages.ros.org/ros2/ubuntu $(source /etc/os-release && echo
$UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

3. Install Humble

First, update the apt repository cache:

```
sudo apt update
```

Then upgrade installed software (ROS2 packages are built on frequently updated Ubuntu systems. Please ensure your system is up to date before installing new packages):

```
sudo apt upgrade
```

Install ROS2 desktop version (recommended), which includes: ROS, RViz, examples and tutorials. The installation command is as follows. To install other versions, replace humble in the command with the corresponding version number:

```
sudo apt install ros-humble-desktop python3-argcomplete
```

Install colcon build tools:

```
sudo apt install python3-colcon-common-extensions
```

4. Configure Environment

- If you are using another version of ROS2, replace the humble version identifier with the corresponding version. For example, for the foxy version, replace humble with foxy in the command.

When running ROS2 programs in the terminal, you need to run the following command to configure the environment:

```
source /opt/ros/humble/setup.bash
```

Every time you open a new terminal, you need to run the above command. Alternatively, you can run the following command to write the environment configuration instruction to the "~/.bashrc" file. Then every time you start a new terminal, you don't need to manually configure the environment:

```
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
```

At this point, ROS2 has been installed and configured.

5. Run Publisher and Subscriber Example Nodes

The ROS2 desktop version comes with some example nodes pre-integrated. We can test if ROS2 is installed successfully by running example nodes.

- Run the publisher node:

```
ros2 run demo_nodes_cpp talker
```

- Run the subscriber node:

```
ros2 run demo_nodes_cpp listener
```

6. Run the TurtleSim Tutorial

The turtlesim simulator is a classic teaching tool in ROS. Through turtlesim, you can quickly understand core concepts in ROS.

- Run the turtle simulator:

```
ros2 run turtlesim turtlesim_node
```

- Run the keyboard control node:

```
ros2 run turtlesim turtle_teleop_key
```

- View velocity topic data:

```
ros2 topic echo /turtle1/cmd_vel
```

7. About Uninstall

After ROS2 is installed, if you want to uninstall ROS2, you can run the following command:

```
sudo apt remove ~nros-humble-* && sudo apt autoremove
```

You can also delete the ROS2 corresponding repository:

```
sudo rm /etc/apt/sources.list.d/ros2.list  
sudo apt update  
sudo apt autoremove
```